

Theorem Queue Proof Ledger

Definitions, assumptions, exit criteria, and kill conditions for the completed one-pass derivation

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Abstract

This compact ledger accompanies the end-to-end closure paper. It records exactly which statements are theorems, which are representation conditions, which are computations, and which remain empirical tests. Its purpose is to prevent later summaries from collapsing distinct claim grades.

Current status: *Use this ledger as the controlling index when revising Volumes 08-10 and the final monograph.*

1. QUEUE SUMMARY

Priority	Theorem target	Exit result	Grade
1	Description-measure invariance	Continuation automorphisms preserve the full continuation profile and its induced measure.	PROVED
2	Integral-record theorem	Additive completed records extend to Λ^* ; saturation and reciprocal pairing yield unimodularity.	PROVED / representation-conditioned
3	Canonical A_2 quotient	Equal reversible zero-sum three-channel closure forces A_2 and covolume $\sqrt{3}$.	PROVED
4	Vacuum-stress variation	Constant scalar record gives $T_{mn} = -\rho g_{mn}$, conservation, and $w = -1$.	PROVED form / model-defined units
5	Scatter kernel	A_2 tight frame fixes restoring coefficient $3/2$ and exponential clock kernel.	PROVED structure / falsifiable model

2. MINIMAL ASSUMPTIONS

ID	Name	Exact statement	Controls
A1	Continuation automorphism	Closed act-record cycle is bijective and preserves admissible continuations in both directions.	Measure theorem
A2	Primitive integral record	One completed record unit exists; records add under gluing and reverse sign under orientation.	Integral dual
A3	Record saturation	Primitive coordinate records form a dual basis and integer record composition is allowed.	Λ^* exhaustion
A4	Reciprocal pairing	Every realised state has a reciprocal record and the state-record map is bijective.	Unimodularity
A5	Full outcome faithfulness	The eight binary outcomes are independent regular record sectors.	Rank eight
A6	Paired positive norm	Self-pairing counts orientation pairs and vanishes only for the null state.	Even positivity
A7	Indecomposable completion	No proper orthogonal subcarrier retains all completion and record sectors.	E8 selection
A8	Equal three-channel closure	Three observer vectors have equal norm, zero sum, and reversible orientation.	A_2 quotient
A9	Minimal genus-zero boundary	The completed observer boundary is connected, orientable, closed, and handle-free.	4π record

A10	Constant scalar persistence	The completed boundary record is spacetime constant in the background solution.	Vacuum stress
A11	Clock gradient flow	Unresolved carrier perturbations relax by the A2 quadratic energy in $d\tau = H dt$.	Scatter kernel

3. THEOREM DEPENDENCY MAP

$$A1 \rightarrow \mu \text{ invariance} \rightarrow J_R J_A = 1$$

$$A2 + A3 + A4 \rightarrow \Lambda \sim \Lambda^*$$

$$A5 + A6 + A7 + \text{self-duality} \rightarrow E8$$

$$A8 \rightarrow A2 \rightarrow \text{covolume} \sqrt{3} \rightarrow (1/\sqrt{3}, \sqrt{3})$$

$$A9 + r^{240} \rightarrow \lambda_{\text{boundary}} = (4\pi/\sqrt{3})r^{240}$$

$$A10 \rightarrow T_{mn} = -\rho g_{mn} \rightarrow w = -1$$

$$A8 + A11 \rightarrow d\xi/d\tau = -(3/2)\xi + \text{forcing}$$

4. WHAT IS NOW CLOSED AND WHAT IS NOT

Statement	Current verdict
There are 241 E8 roots	FALSE. There are 240 roots. The +1 is a whole-boundary operation, not a root.
The exponent must be 241	NOT REQUIRED. The snowflake readout acts on r^{240} by a geometric functional.
A unique forty-cell partition must be proved	NO. The quotient type A2 and its covolume are cover-independent; an exhaustive atlas must merely exist.
Description measure needs a conservation postulate	NO within the continuation model. Invariance follows from graph automorphism.
Records-as-functionals automatically imply self-duality	NO. A reciprocal saturated state-record pairing is additionally required.
The vacuum stress form is derived	YES under constant scalar, locality, and diffeomorphism invariance.
The physical theory is independently established	NO. Representation membership and empirical tests remain open.

5. FROZEN PREDICTIONS AFTER THE QUEUE

ID	Prediction	Primary kill condition
P ₁	$w_0 = -1$, $w_a = 0$ exactly while the completed scalar record persists.	Robust evolving dark energy across independent probes.
P _{2-prime}	$a(z) = \sqrt{3} c H(z) \Delta t_B$; $a_0 = 1.1180 \times 10^{-10} \text{ m s}^{-2}$.	Independent catalogues outside frozen tolerance or failure of $H(z)$ tracking.
P ₃	Mean acceleration scale is environment-independent at fixed epoch.	Repeatable matched environment offset.
S ₁	Residual memory decays with exponent $3/2$ in cosmological clock time.	Held-out free-exponent kernel excludes $3/2$ or no relaxation signal exists.

6. NEXT OPERATIONS - NO NEW THEOREM BRANCHES

1. Audit the physical act-record map against A_1 rather than adding another measure principle.
2. Extract the primitive record basis and reciprocal pairing from the current loop algebra notation and test A_2 - A_4 line by line.
3. Verify the rank-eight regular representation and indecomposability conditions before using the word forced for E8.
4. Rerun the finite forty-cell certificate independently; treat alternative covers as atlas gauge unless they change the local Gram determinant.
5. Implement the fixed $3/2$ scatter kernel in the preregistered residual pipeline without refitting the mean amplitude.
6. Keep all failed membership tests and negative empirical results in the public ledger.